

PRAKTIKUM SISTEM OPERASI

MODUL 7

Semaphore



Oleh:

Jauhar Fajar Zuhair

2311104072

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

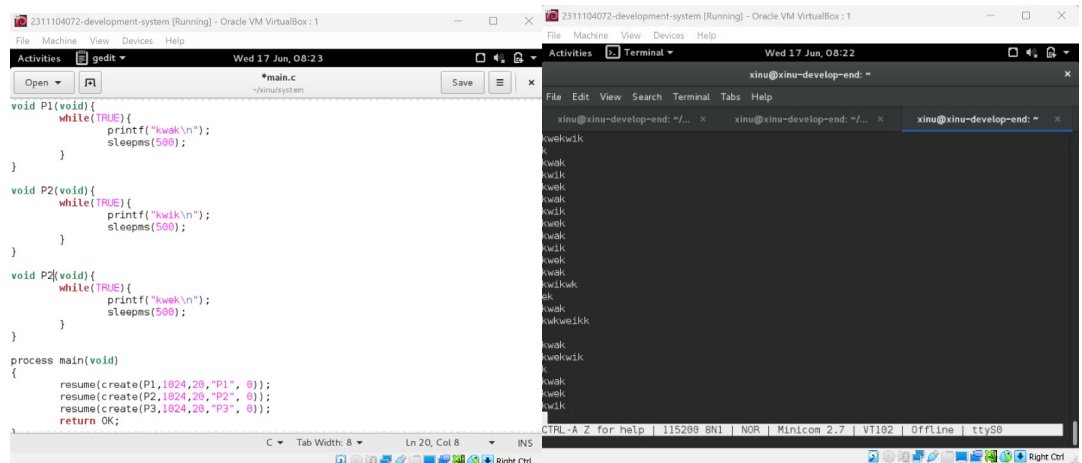
DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

JURNAL

1. Soal 1



The screenshot shows a virtual machine window titled "2311104072-development-system [Running] - Oracle VM VirtualBox : 1". It contains two windows: a code editor and a terminal.

The code editor window shows a C program named `main.c` with the following code:

```
void P1(void){
    while(TRUE){
        printf("kwak\n");
        sleeps(500);
    }
}

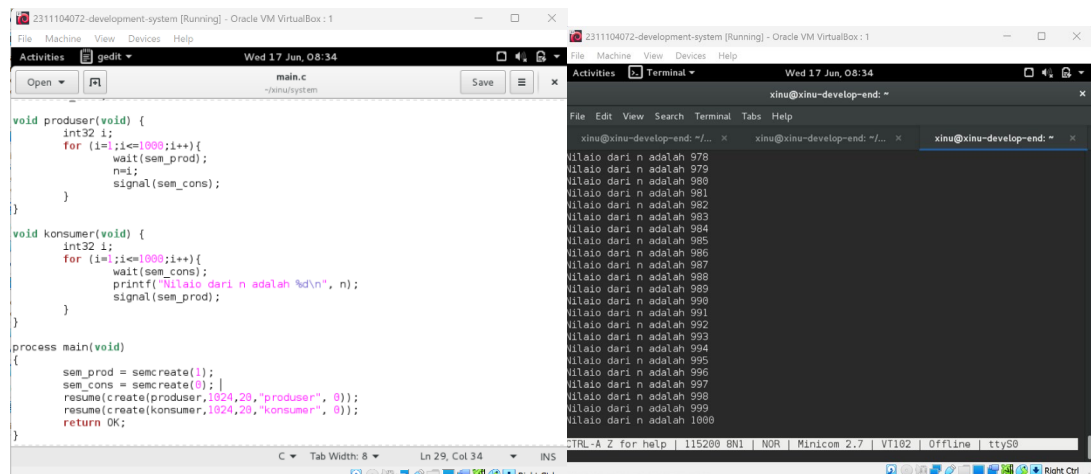
void P2(void){
    while(TRUE){
        printf("kwak\n");
        sleeps(500);
    }
}

void P3(void){
    while(TRUE){
        printf("kwak\n");
        sleeps(500);
    }
}

process main(void)
{
    resume(create(P1,1024,20,"P1", 0));
    resume(create(P2,1024,20,"P2", 0));
    resume(create(P3,1024,20,"P3", 0));
    return OK;
}
```

The terminal window shows the output of the program, which consists of multiple lines of "kwak" printed in a staggered fashion, indicating concurrent execution of the three processes.

2. Soal 2



The screenshot shows a virtual machine window titled "2311104072-development-system [Running] - Oracle VM VirtualBox : 1". It contains two windows: a code editor and a terminal.

The code editor window shows a C program named `main.c` with the following code:

```
void producer(void) {
    int32 i;
    for (i=1;i<=1000;i++){
        wait(sem_prod);
        n=i;
        signal(sem_cons);
    }
}

void konsumen(void) {
    int32 i;
    for (i=1;i<=1000;i++){
        wait(sem_cons);
        printf("Nilai dari n adalah %d\n", n);
        signal(sem_prod);
    }
}

process main(void)
{
    sem_prod = semcreate(1);
    sem_cons = semcreate(0);
    resume(create(producer,1024,20,"producer", 0));
    resume(create(konsumer,1024,20,"konsumer", 0));
    return OK;
}
```

The terminal window shows the output of the program, which consists of multiple lines of "Nilai dari n adalah" followed by a number, indicating the execution of the consumer process.